BRUNVOLL

Technical Report

Propulsion Optimiser – Fuel Saving Potential

Brunvoll Document No. XXXXXX(Pre)

April 14, 2026

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1 Introduction

This report presents fuel saving results from Brunvoll's voyage fuel simulation tool, developed to quantify the potential of the **Propulsion Optimiser** – our smart propeller control system. The analysis also evaluates the combined benefit when the Propulsion Optimiser is paired with **Flettner rotor wind-assist**, as the two technologies are complementary: the Optimiser improves engine efficiency at every operating point, while the Flettner rotor reduces the thrust demand itself by harnessing wind energy. Together, they address different parts of the propulsion energy chain and their savings add up.

- **The Propulsion Optimiser** (referred to as "Optimiser" in the rest of this report) – finds the most fuel-efficient propeller pitch/RPM combination for any given condition, replacing the standard (factory-default) propeller control schedule.
- **Flettner rotor wind-assist** – a spinning cylinder mounted on deck that generates forward thrust via the Magnus effect, reducing the load on the main engine.

To evaluate these technologies under realistic conditions, we simulate a full year (2024) of repeated voyages on a representative North Sea route (Rotterdam–Gothenburg, 509 nm), using actual historical weather data. Each voyage experiences the real waves and winds that occurred on that date, giving a comprehensive picture of fuel savings across all seasons and weather conditions.

1.1 What We Compare

The simulation runs every voyage under four configurations:

Configuration	Description
Standard control, no rotor	Factory-default propeller schedule, no wind-assist – the baseline
Standard control + Flettner rotor	Factory schedule with wind-assist thrust from the rotor
Optimiser, no rotor	Fuel-optimised pitch/RPM selection, no wind-assist
Optimiser + Flettner rotor	Fuel-optimised pitch/RPM combined with wind-assist – the full-benefit case

Fuel savings are always measured relative to the baseline (standard control, no rotor), and are decomposed into:

- **Propeller optimisation saving** – the Optimiser's benefit from smarter pitch/RPM selection, independent of weather.
- **Wind-assist saving** – the additional benefit from the Flettner rotor, which grows with wind speed.

2 How the Simulation Works

2.1 Hull Resistance and Propulsion

The simulation models the vessel's propulsion chain from hull resistance through to engine fuel consumption:

- **Calm-water resistance** is based on the vessel's model test trial prediction (test 25-0461, draught 7.600 m), covering speeds from 8 to 15.0 knots.

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- **Wave added resistance** is computed using hydrodynamic transfer functions specific to the hull shape, integrated over the wave spectrum at each time step. In simple terms: the tool calculates how much extra drag the waves create based on their height, period, and direction relative to the vessel.
- **Wind resistance on the hull** follows the established Blendermann (1994) method, accounting for wind speed and direction relative to the vessel. Headwinds add resistance; following winds reduce it.
- **Hull and propeller fouling** can be included as time-dependent roughness growth. The baseline results shown here assume clean hull and propeller conditions.

2.2 Flettner Rotor Wind-Assist

The Flettner rotor is a spinning vertical cylinder ($28\,\mathrm{m}$ tall, $4\,\mathrm{m}$ diameter) that exploits the Magnus effect: when wind flows past the spinning rotor, it generates a sideways force – much like a spinning ball curves in flight. The forward-pointing component of this force acts as free thrust that supplements the propeller.

The key equation governing the rotor's thrust is:

$$F_{\text{rotor}} = C_L(\text{SR}) \cdot \tfrac{1}{2}\,\rho_{\text{air}}\,V_{\text{app}}^2 \cdot D \cdot H \tag{1}$$

where C_L is the lift coefficient (dependent on the spin ratio SR – how fast the rotor spins relative to the wind speed), V_{app} is the apparent wind speed, and $D \times H$ is the rotor's projected area. The rotor's own motor power consumption is subtracted to give the net fuel benefit.

The rotor produces the most benefit in beam winds (wind from the side) and provides diminishing returns in headwinds or following winds. Its effectiveness increases with wind speed.

2.3 Standard vs Optimiser Propeller Control

A controllable-pitch propeller (CPP) has two adjustment parameters: blade pitch and shaft RPM. The **standard control** uses a fixed schedule that links these two parameters together – a single lever controls both, following a pre-set curve.

The **Optimiser** breaks this rigid link. For any given thrust requirement, it searches the full range of pitch and RPM combinations to find the one that minimises fuel consumption, while respecting the engine's operating limits. The saving comes from running the engine at a more efficient load point – typically lower RPM and higher pitch than the standard schedule would dictate.

2.4 Weather Data

The simulation uses wave and wind data from NORA3, the third-generation Norwegian Reanalysis Archive produced by MET Norway. This is a high-quality historical weather reconstruction that provides complete, gap-free wave and wind fields across the North Sea at approximately $3\,\mathrm{km}$ spatial resolution and hourly time steps.

Unlike isolated buoy measurements, this dataset covers the entire route with consistent quality, ensuring that every simulated voyage encounters realistic weather conditions. The archive spans from 1993 to the present, enabling multi-year studies if needed.

3 Simulation Setup

The study simulates a full year of round-trip voyages on the route shown in Figure 3-1.

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Parameter	Value	Unit
Route	509	nm (one-way)
Transit speed	10	kn
Transit time (one-way)	50.9	hours
Mode	Round-trip	
Flettner rotor	28 m tall, 4 m dia	
Hull condition	Clean (no fouling)	
Port/idle time	15	%
Fuel price (MGO)	650	EUR/tonne
Propeller diameter	4.80	m
Main engine	MAN L27/38	2920 kW

Each voyage departs daily, sails the outbound leg, and returns immediately – experiencing different weather on each leg. Round-trip simulation eliminates any directional wind bias (e.g. prevailing westerlies favouring one direction). With 15% assumed port/idle time, the vessel completes approximately 73 round-trip voyages per year.

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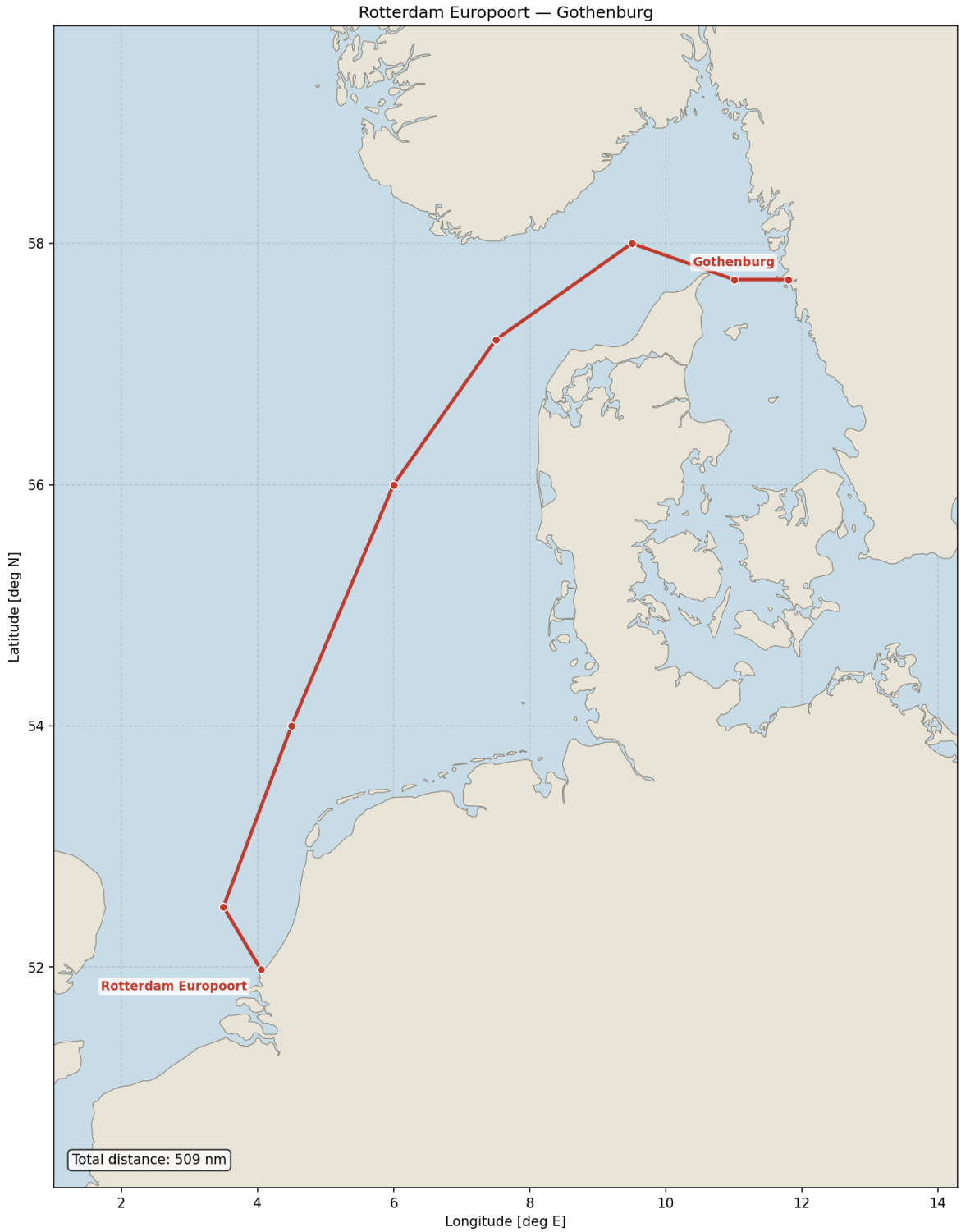

Figure 3-1: Rotterdam–Gothenburg route (509 nm one-way) with waypoints.

4 Results

362 round-trip voyages were successfully simulated across the full year 2024.

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4.1 Per-Voyage Fuel Consumption

Metric	Mean	Median	P10	P90	Min	Max
Standard, no rotor [kg/voyage]	17378	17241	15910	19444	10816	23079
Optimiser + rotor [kg/voyage]	15120	15147	12949	17443	8107	21971
Propeller optimisation saving [kg]	851	883	729	936	409	981
Wind-assist saving [kg]	1407	1245	360	2660	3	4847
Propeller optimisation saving	4.9%	5.1%	3.9%	5.7%	2.6%	6.4%
Wind-assist saving	8.3%	7.1%	2.0%	16.6%	0.0%	28.6%

Key observations:

- The **propeller optimisation** delivers a consistent 3–6% saving on every voyage, regardless of weather.
- The **wind-assist saving** varies from near zero in calm conditions to over 28% on the windiest voyages, reflecting its weather dependence.
- On the single best voyage, the Flettner rotor alone saved 4.9 tonnes of fuel.

The savings are additive: the combined benefit equals the propeller optimisation saving plus the wind-assist saving.

Figure 4-1 shows how the savings evolve over the year, alongside the sea conditions that drive them.

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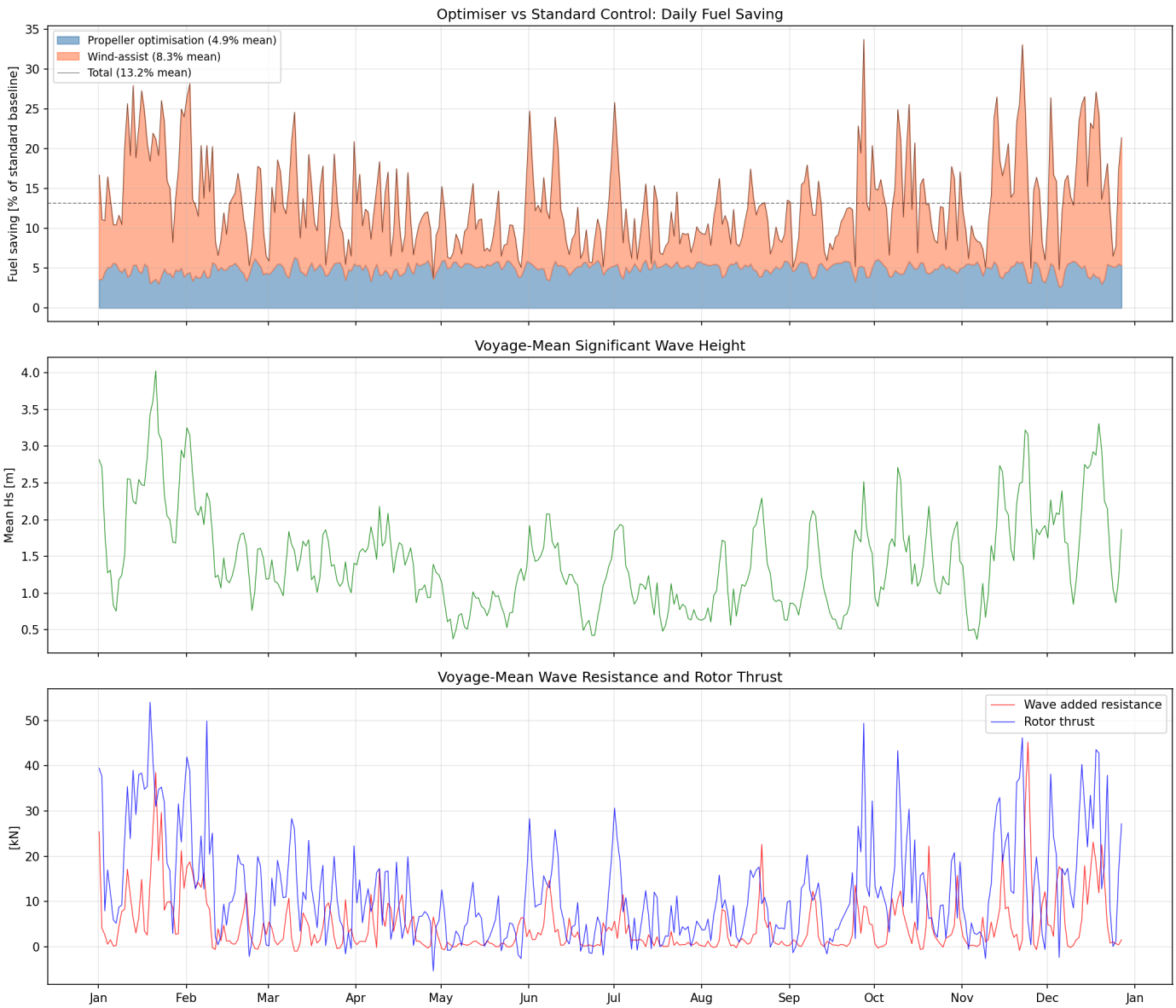

Figure 4-1: Annual results per voyage. Top: propeller optimisation and wind-assist fuel savings (stacked). Middle: significant wave height. Bottom: wave resistance, wind resistance, and Flettner rotor thrust.

4.2 Weather Correlation

Figure 4-2 shows how the fuel saving on each voyage correlates with wave height and wind speed. The trend is clear: windier conditions mean larger savings from the Flettner rotor.

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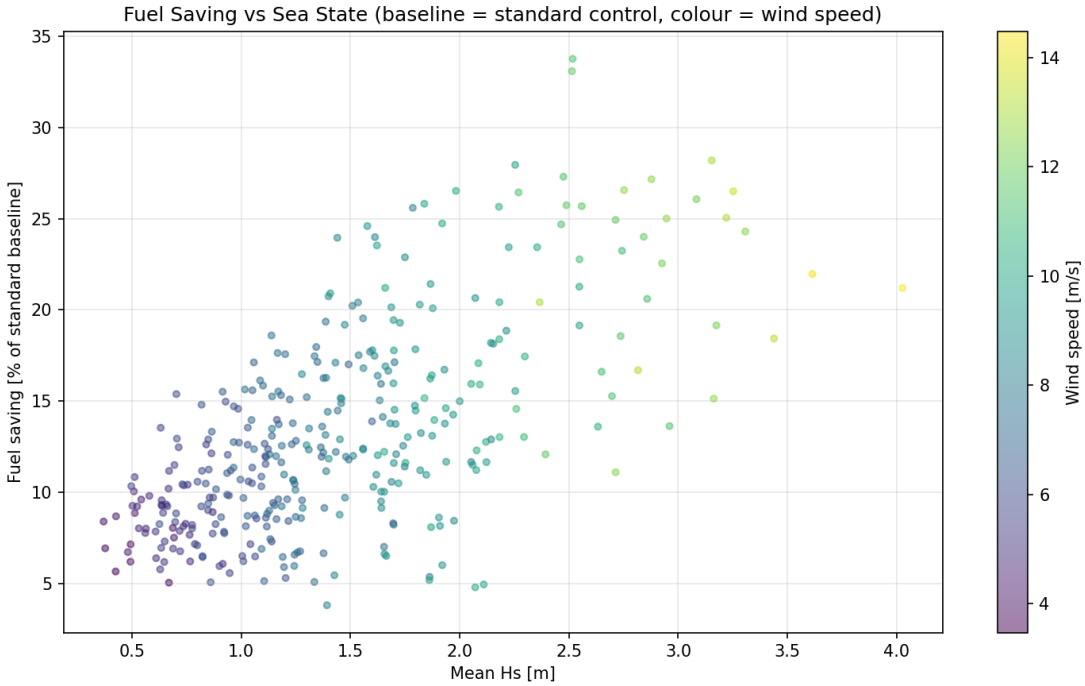

Figure 4-2: Per-voyage fuel saving vs mean wave height, coloured by mean wind speed. Windier conditions produce larger savings through the Flettner rotor.

4.3 Annualised Fuel Consumption

Configuration	Fuel [tonnes/year]
Standard control, no rotor	1270.8
Standard control + Flettner rotor	1167.4
Optimiser, no rotor	1208.5
Optimiser + Flettner rotor	1105.6

Saving component	Tonnes/year	Percentage
Propeller optimisation	62.3	4.9%
Wind-assist (Flettner)	102.9	8.1%
Combined	**165.2**	**13.0%**

Figure 4-3 shows the distribution of per-voyage savings. The propeller optimisation is tightly clustered (consistent benefit), while the wind-assist saving shows a wider spread reflecting its dependence on weather.

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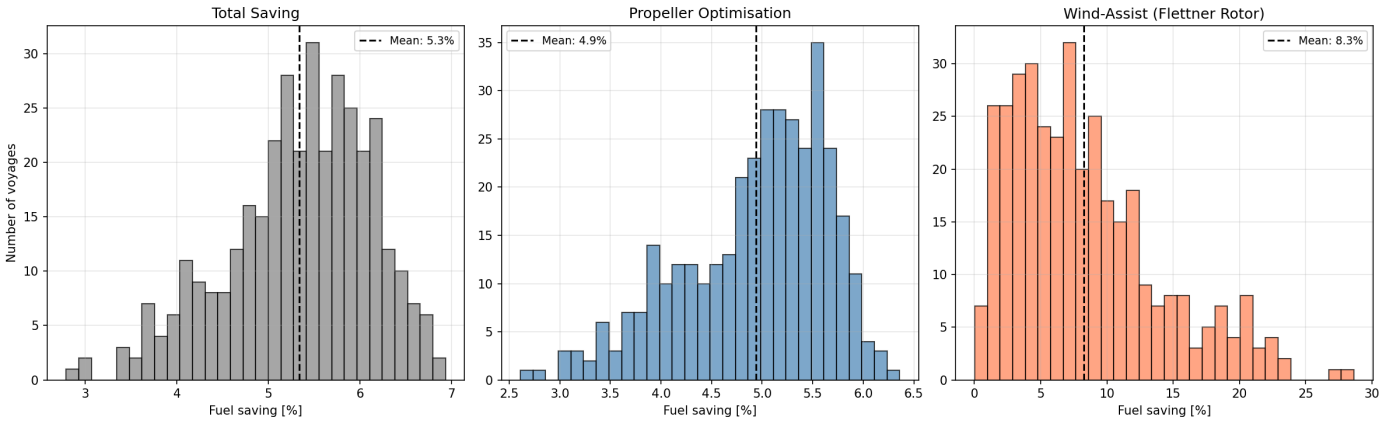

Figure 4-3: Distribution of per-voyage fuel savings. Left: total saving. Centre: propeller optimisation component. Right: wind-assist component.

4.4 Annual Cost Savings

At a fuel price of €650/tonne (MGO, typical for ECA zones):

Configuration	Annual fuel cost
Standard control, no rotor	€826 036
Standard control + Flettner rotor	€758 839
Optimiser, no rotor	€785 543
Optimiser + Flettner rotor	€718 687

Saving component	€/year	Percentage
Propeller optimisation	€40 494	4.9%
Wind-assist (Flettner)	€66 855	8.1%
Combined	**€107 349**	**13.0%**

4.5 Seasonal Variation

Season	Voyages	Std [kg/voy]	Opt+Fl [kg/voy]	Prop. opt.	Wind-assist
Q1 (Jan–Mar)	91	17 420	14 835	4.7%	10.4%
Q2 (Apr–Jun)	91	17 519	15 607	5.1%	6.0%
Q3 (Jul–Sep)	92	17 348	15 377	5.1%	6.3%
Q4 (Oct–Dec)	88	17 220	14 642	4.9%	10.4%

The Flettner rotor benefit roughly doubles in winter (Q1, Q4) compared to summer, reflecting the stronger and more consistent North Sea winds. The propeller optimisation saving remains stable at 4.7–5.1% year-round, as it depends on engine efficiency rather than weather.

4.6 Standard Control Limitations

The standard propeller control schedule was not designed to handle all possible sea conditions. In 7% of simulated voyages, the standard schedule drives the engine beyond its power limits in at least one hour – a situation where the operator would need to manually reduce speed.

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The Optimiser avoids this entirely by choosing pitch/RPM combinations that respect the engine's full operating envelope, maintaining the desired speed even in heavy weather.

5 Effect of Shaft Generator Operation

5.1 Why It Matters

The results above assume the vessel operates without a shaft generator (SG). Many vessels use an SG to supply electrical power from the main engine, avoiding the need for a separate diesel generator during transit. However, the SG introduces two constraints that affect propulsion efficiency:

1. **Additional engine load:** The SG extracts electrical power (assumed $150\,\mathrm{kW}$ average) from the main engine shaft, increasing fuel consumption.
2. **Minimum RPM constraint:** The SG must maintain its output frequency within $48\,\mathrm{Hz}$ to $60\,\mathrm{Hz}$, which sets a minimum engine RPM of approximately $640\,\mathrm{RPM}$ (shaft RPM $\geq 94.1\,\mathrm{RPM}$). Neither the standard schedule nor the Optimiser can operate below this RPM while the SG is online.

5.2 Results with Shaft Generator Active

Configuration	Fuel [tonnes/year]
Standard control, no rotor	1535.7
Standard control + Flettner rotor	1452.3
Optimiser, no rotor	1535.4
Optimiser + Flettner rotor	1453.3

Saving component	Tonnes/year	Percentage
Propeller optimisation	0.3	0.0%
Wind-assist (Flettner)	82.1	5.3%
Combined	**82.4**	**5.4%**

At MGO €650/tonne, the annual saving with SG active is €54 000/year (5.4%).

5.3 Effect on Propeller Optimisation

At $10\,\mathrm{kn}$ transit speed, the calm-water thrust demand is approximately $133\,\mathrm{kN}$, requiring shaft power of approximately $767\,\mathrm{kW}$. With the SG RPM floor at $94.1\,\mathrm{RPM}$ (shaft), both the standard schedule and the Optimiser are pinned at essentially the same operating point: the RPM floor removes the Optimiser's primary strategy of running at lower RPM and higher pitch. As a result, the **propeller optimisation saving drops to 0.0%** with the SG active — all of the combined saving comes from the Flettner rotor.

At higher transit speeds, where the thrust demand pushes the engine above the SG RPM floor, the Optimiser would regain its pitch/RPM freedom and deliver meaningful savings.

5.4 Flettner Rotor Remains Effective

The Flettner rotor delivers the same absolute thrust reduction regardless of SG status – wind physics are unchanged. The relative saving appears smaller (5.3% vs 8.1%) because the SG-loaded baseline consumption is higher ($1536\,\mathrm{t/yr}$ vs $1271\,\mathrm{t/yr}$).

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Season	Voyages	Std [kg/voy]	Opt+Fl [kg/voy]	Prop. opt.	Wind-assist
Q1 (Jan–Mar)	91	20 991	19 554	0.1%	6.9%
Q2 (Apr–Jun)	91	21 148	20 330	-0.0%	3.9%
Q3 (Jul–Sep)	92	20 990	20 135	-0.1%	4.1%
Q4 (Oct–Dec)	88	20 870	19 460	0.0%	6.8%

5.5 Side-by-Side Summary

Metric	SG offline	SG online (150 kW)
Baseline fuel [t/yr]	1270.8	1535.7
Propeller optimisation saving	4.9% (62 t/yr)	0.0% (0.3 t/yr)
Wind-assist saving	8.1% (103 t/yr)	5.3% (82 t/yr)
Combined saving	13.0% (165 t/yr)	5.4% (82 t/yr)
Annual cost saving	€107k	€54k

The SG operating mode increases total fuel consumption by approximately 21% due to the parasitic load. Critically, the SG RPM floor constraint eliminates the Optimiser's pitch/RPM saving at $10\,\mathrm{kn}$, as both the standard schedule and the Optimiser are pinned at the same operating point. The Flettner rotor benefit is preserved in absolute terms.

At higher transit speeds, the propeller optimisation saving is expected to show similar or improved performance under SG constraints, as the thrust demand pushes the engine further above the SG RPM floor:

Transit speed	Propeller opt.	Wind-assist	Combined
$10\,\mathrm{kn}$	0.0%	5.3%	5.4%

5.6 Shaft Generator vs Auxiliary Genset

An important question is whether the shaft generator is the most fuel-efficient way to supply $150\,\mathrm{kW}$ of electrical power at $10\,\mathrm{kn}$.

With the SG, the baseline fuel rises from $1271\,\mathrm{t/yr}$ to $1536\,\mathrm{t/yr}$ — an increase of $265\,\mathrm{t/yr}$ for $150\,\mathrm{kW}$ over $7439\,\mathrm{h/yr}$. This corresponds to an *effective* specific fuel consumption of $237\,\mathrm{g/kWh}$, notably above the 210 –-$230\,\mathrm{g/kWh}$ of a typical auxiliary diesel genset.

At first glance $237\,\mathrm{g/kWh}$ may seem surprisingly high: the main engine's own SFOC in its sweet spot is only 181 –-$185\,\mathrm{g/kWh}$. The explanation is that the effective SFOC includes an *opportunity cost*. Without the SG the Optimiser selects ~$75\,\mathrm{RPM}$ shaft ($510\,\mathrm{RPM}$ engine) at $10\,\mathrm{kn}$, deep in the fuel map's efficient region. With the SG engaged, the minimum-frequency requirement forces the engine to $\geq 640\,\mathrm{RPM}$ ($94\,\mathrm{RPM}$ shaft), completely eliminating the Optimiser's pitch/RPM freedom. The extra fuel therefore has two components: (i) the raw energy for $150\,\mathrm{kW}$, and (ii) the efficiency loss from operating at a constrained, less optimal RPM — plus the total loss of the Optimiser's 4.9% pitch/RPM saving. Together they yield the $237\,\mathrm{g/kWh}$ effective rate.

At a typical genset SFOC of $215\,\mathrm{g/kWh}$, a separate auxiliary genset would save approximately $25\,\mathrm{t/yr}$ (€16 000/year) in fuel compared to the shaft generator. This saving is in addition to the Optimiser and Flettner benefits.

Figure 5-1 illustrates this contrast across the full year.

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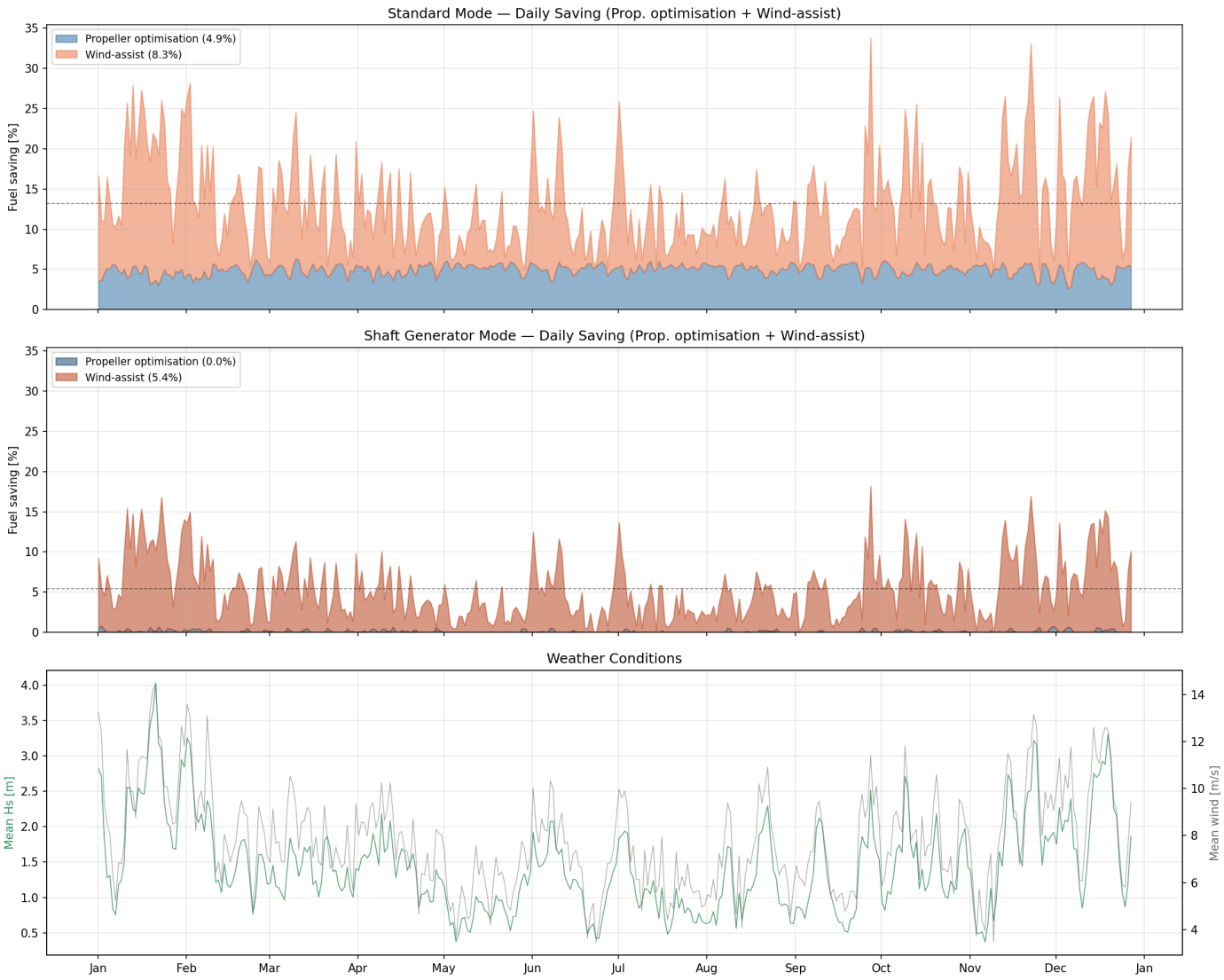

Figure 5-1: Daily fuel savings: standard mode (top) vs shaft generator mode (middle), with weather conditions (bottom). In standard mode, the propeller optimisation saving (blue) is consistent; in SG mode, the RPM floor constraint eliminates it. The wind-assist saving (orange) persists in both modes. Dashed lines show annual means.

6 Conclusions

The simulation of 362 round-trip voyages on the Rotterdam–Gothenburg route using actual 2024 weather data demonstrates fuel saving potential from two complementary technologies. The achievable savings depend on whether the shaft generator is in use:

Without shaft generator (SG offline):

1. **The Optimiser** saves **4.9%** of fuel (62 tonnes/year, €40 000/year). This is a consistent, weather-independent benefit delivered on every voyage by running the engine at a more efficient operating point.
2. **Flettner rotor wind-assist** saves an additional **8.1%** (103 tonnes/year, €67 000/year), varying with wind conditions – roughly 6% in summer and 10% in winter.

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3. **Combined saving: 13.0%** (165 tonnes/year, €107 000/year).

With shaft generator (SG online, 150 kW, 48–60 Hz):

4. The SG increases baseline fuel by approximately 21% due to the parasitic load. At $10\,\mathrm{kn}$, the SG RPM floor constraint pins both the standard schedule and the Optimiser at the same operating point, **eliminating the pitch/RPM saving entirely** (0.0%).
5. The **Flettner rotor remains effective at 5.3%** (82 tonnes/year). The lower percentage reflects the higher baseline consumption from the SG parasitic load, not a reduction in actual wind-assist performance.
6. **Combined saving with SG: 5.4%** (82 tonnes/year, €54 000/year).
7. **Shaft generator vs auxiliary genset:** At $10\,\mathrm{kn}$, the effective fuel cost of generating $150\,\mathrm{kW}$ via the shaft generator is $237\,\mathrm{g/kWh}$ — notably above a typical auxiliary genset at $215\,\mathrm{g/kWh}$. A separate genset would save approximately €16 000/year.

In both modes:

7. The Optimiser eliminates engine overload situations that occur with the standard schedule in heavy weather, improving operational reliability.

These results assume clean hull and propeller surfaces. Fouling (biological growth on the hull and propeller) increases resistance over time, which would increase the absolute fuel savings from both technologies while also increasing total consumption. The simulation tool supports fouling scenarios and can provide estimates for any assumed maintenance interval.

7 Appendix: Methodology Notes

The physical models underlying the simulation are based on established naval architecture methods:

- **Hull resistance:** Interpolated from vessel-specific model test trial prediction.
- **Wave added resistance:** Strip-theory hydrodynamic transfer functions (PdStrip) integrated over JONSWAP wave spectra.
- **Wind resistance:** Blendermann (1994) method for general cargo vessels.
- **Flettner rotor:** Lift/drag coefficients from Bordogna et al. (2019) with aspect-ratio correction; rotor drive power from Theodorsen & Regier (1944).
- **Fouling:** Hull roughness model of Townsin & Dey (2003); propeller blade roughness from Mosaad (1986).
- **Weather data:** NORA3 hindcast archive (MET Norway), $3\,\mathrm{km}$ resolution, hourly time steps.
- **Propulsion model:** C-series propeller model with full pitch/RPM optimisation against the MAN L27/38 engine envelope.

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